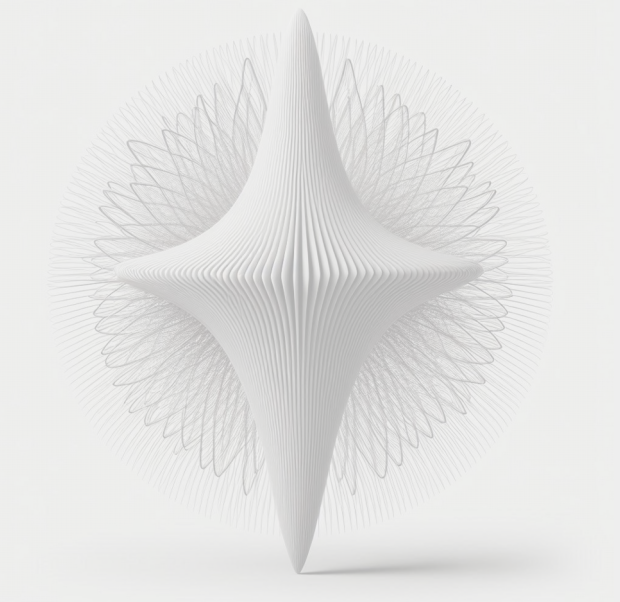


Signals and Systems

Lakshmi Narasimhan
IIT Madras

Disclaimer: There could be errors and typos in these slides. Use with caution.

Fourier Analysis

Eigenfunctions of LTI Systems

- Convolution is complicated! Can we simplify?
  - Exponentials are eigenfunctions

$$h(t) * e^{j\omega t} = \int h(\tau) e^{j\omega(t-\tau)} d\tau = e^{j\omega t}\lambda; \qquad \lambda = \int h(\tau) e^{-j\omega\tau} d\tau$$

  - Complex exponentials are orthogonal and bounded
- What if the inputs were only eigenfunctions (or their combinations)?

2D Vectors Analogy

- Representation of a vector ($\mathbf{z}$) in terms of its bases ($\mathbf{x}$, $\mathbf{y}$)
  - Any 2D vector [a, b] can be expressed as a linear combination of [1, 0] and [0, 1]
    - $\mathbf{z} = a\mathbf{x} + b\mathbf{y}$
  - Possible choices of bases are many; e.g., [1, 1] and [1, -1]
  - When ($\mathbf{x}$, $\mathbf{y}$) are orthogonal ($\mathbf{x}^T\mathbf{y} = 0$): $a = \mathbf{x}^T\mathbf{z}/\mathbf{x}^T\mathbf{x}$ and $b = \mathbf{y}^T\mathbf{z}/\mathbf{y}^T\mathbf{y}$
- Orthogonal basis are most useful
- Generalizing this notion to higher dimensions, $\mathbf{z} = \Sigma_i a_i \mathbf{x}_i$, $a_i = \mathbf{x}_i^T\mathbf{z}/\mathbf{x}_i^T\mathbf{x}_i$

Complex Sinusoids as Orthogonal Bases

- In convolution, $\delta(t - \tau)$ is the basis (for different values of τ); $x(\tau)$ = inner product
- **Fourier Analysis**: Using eigenfunctions as orthogonal basis for system analysis
- Complex sinusoids (phasors) as Orthogonal bases (eigenfunctions of LTI systems)
  - $x(t) = \Sigma_i c_i b_i(t)$, where $b_i(t)$ are orthogonal basis (phasors of different frequencies)
  - As before, can we compute coefficients c_i as $\langle x(t), b_i(t)\rangle / \langle b_i(t), b_i(t)\rangle$?
  - How many $b_i(t)$ do we need to represent any signal $x(t)$?

How Many Bases are Needed?

- The error (signal) $e(t) = x(t) - \Sigma_i c_i b_i(t)$ should be minimized
  - Magnitude of e(t) should be minimized, i.e., energy E_e should be minimized

$$\arg\min_{c_i} E_e = \arg\min_{c_i} \int |x(t) - \sum_i c_i b_i(t)|^2 dt = \frac{< x(t), b_i(t) >}{< b_i(t), b_i(t) >} = \frac{\int x(t) b_i^*(t) dt}{E_{b_i}}$$

  - Substituting for c_i in e(t), we get $E_e = E_x - \Sigma_i |c_i|^2 E_{bi}$
    - Note that $E_x \geq 0$, $|c_i|^2 E_{bi} \geq 0$ and $E_x \geq \Sigma_i |c_i|^2 E_{bi}$
    - Thus, E_e can be driven to zero by considering infinitely many bases $b_i(t)$
  - General Parseval's theorem: sum of energy of orthogonal signals: $E_x = \Sigma_i |c_i|^2 E_{bi}$

Fourier Series

Fourier Analysis with Phasors

- Complex sinusoids are periodic e^{jwt} and have infinite energy – problem
- Consider only periodic signals
- Let the basis be phasors of frequencies: kw, k = ± 1, ± 2, ± 3, ± 4, ...
  - Periods: $2\pi/w$, $2\pi/2w$, $2\pi/3w$, $2\pi/4w$, … = T, $T/2$, $T/3$, $T/4$, ...
  - This repeats after T: focus only on a finite duration – making energy finite
    - Represent any periodic $x(t)$ within the period T: $x(t) = \Sigma_i c_i b_i(t) \mid_{t = -T/2 \text{ to } T/2}$
- Inner product of finite duration signal: $\int_{-T/2}^{T/2} x(t) b_i^*(t) dt$

Harmonics

- Choice of Harmonics: $k\omega$, $k = 0, \pm 1, \pm 2, \pm 3, \pm 4, \ldots$
  - When periodic signals are added, the sum has a
    - Period that is least common multiple of components
    - Frequency that is greatest common divisor
    - Cannot represent aperiodic or signals that have frequency other than $k\omega$
- If we can decompose the periodic signal $x(t) = \Sigma_i c_i b_i(t)$ in one period, then all the other periods also follow the same decomposition, i.e., $x(t) = \Sigma_i c_i b_i(t) \mid_{t = (a, a + kT)}$

Fourier Series

- Representing a periodic signal of period T with phasors:

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega t}; \qquad \omega = \frac{2\pi}{T}$$

$$c_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jk\omega t} dt$$

  - $x(t) = \Sigma_k c_k b_k(t)$, where $b_k(t)$ are phasors and $c_k = \langle x(t), b_i(t) \rangle / \langle b_i(t), b_i(t) \rangle$
  - In duration T, E_{bi} for a phasor is $\langle b_i(t), b_i(t) \rangle = T$
  - Fourier series representation of (a CT signal) x(t) → c_k (a DT signal)

Fourier Series - Spectrum

- Fourier series coefficient (c_k) is the spectrum of that periodic signal
  - Gives the frequency components of $x(t)$
  - c_k gives the magnitude & phase of each phasor present in $x(t)$
- Bandwidth: (of real signals) Kw: Smallest K s.t. $c_k = 0$ for $|k| > K$
  - Kw rad/s or K/T Hz
  - Other measures of bandwidth = 3dB bandwidth

The Return of Phasors

- Can we represent any periodic signal using only phasors?
  - Signal can be discontinuous but phasors are continuous!
- Matlab demo : Gibbs phenomenon
  - Some times infinitely many phasors are required to represent a signal
  - Occurs due to a discontinuity
    - Rate of change of the signal (frequency) is almost infinity at discontinuities
- At each time instant, the signal is represented as a series: $x(a) = \Sigma_n c_n e^{jnwa}$

Convergence of Fourier Series

- Will the Fourier series of some $z(t)$ always converge?
  - Dirichlet conditions: In a finite period T,
    - Condition 1: $z(t)$ must be absolutely integrable in a period
    - Condition 2: $z(t)$ must have only a finite number of maxima and minima
    - Condition 3: $z(t)$ must have only a finite number of discontinuities
- At the point of discontinuities, FS converges to the mean of left & right limits (MVT)
  - Gibbs phenomenon

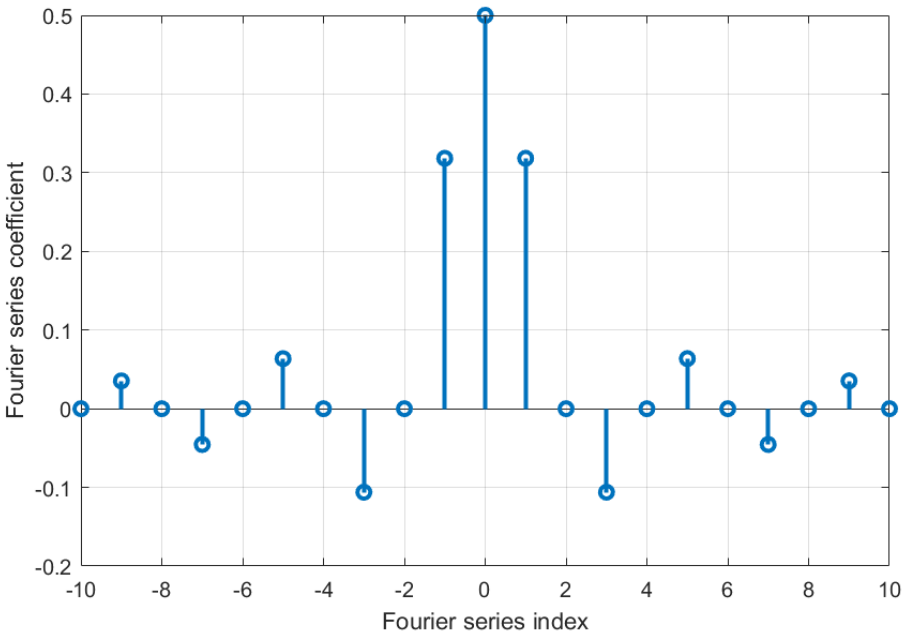
Fourier Series - Example

- Find the Fourier series of a periodic rectangular/square pulse

Fourier Series - Example

- Find the Fourier series of a periodic rectangular/square pulse

$$
\begin{aligned}
c_k &= \frac{1}{T}\frac{2\sin k\omega T/4}{k\omega} \\
&= \frac{1}{2T}T\,sinc\frac{k\omega T/4}{\pi} \\
&= \frac{1}{2}sinc(k/2)
\end{aligned}
$$



- As T increases, the bandwidth decreases, and vice versa

Fourier Series - Properties

$z(t)$ – Time domain	a_k – Fourier coefficients
Even	Even
Odd	Odd
Conj. Sym.	Real
Conj. Asym.	Imaginary
Real	Conj. Sym.
Imaginary	Conj. Asym.

Fourier Series - Properties

$z(t)$ – Time domain	a_k – Fourier coefficients	Bandwidth
$z^*(t)$	a_{-k}^*	No change
$z(-t)$	a_{-k}	No change
$z(t - s)$	$a_k e^{-jkws}$ (Only phase changes)	No change
$z(ct)$	$a_{k/c}$ (or coefficients are the same but fundamental frequency changes to wc)	c > 1: expands c < 1: shrinks

Fourier Series - Properties

- Parseval's theorem, $E_z = \Sigma_i |a_i|^2 E_{xi}$
  - For periodic signals and given FS:
    - Energy in one period: $E_{z,T} = T \Sigma_k |a_k|^2$
    - Power of the signal: $P_z = E_{z,T}/T = \Sigma_k |a_k|^2$

Fourier Series - Properties

- For two periodic signals $z(t)$ and $v(t)$ of same period, $z(t)\ v(t) \rightarrow a_k * b_n$
  - Time multiplication results in spectral convolution

Fourier Series – Simplifying Convolution

- For a system with impulse response $h(t)$ (need not be periodic) and periodic input $z(t)$:

$$z(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega t}; \qquad \omega = \frac{2\pi}{T} \qquad h(t) * e^{j\omega t} = e^{j\omega t}\lambda_\omega; \qquad \lambda_\omega = \int h(\tau)e^{-j\omega\tau}d\tau$$

$$a_k = \frac{1}{T}\int_{-T/2}^{T/2} z(t)e^{-jk\omega t}dt \qquad z(t) * h(t) = \sum_{k=-\infty}^{\infty} a_k \lambda_{k\omega} e^{jk\omega t}$$

  - Output is another periodic signal with FS coefficients = $a_k\lambda_{kw}$
    - Bandwidth reduces or remains the same
  - λ_{kw} is the frequency response of the LTI Systems

Periodic Convolution

- For a system with periodic impulse response $h(t)$ and periodic input $z(t)$:

$$z(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega t}; \qquad \omega = \frac{2\pi}{T} \qquad\qquad h(t) = \sum_{m=-\infty}^{\infty} b_m e^{jm\omega t}; \qquad \omega = \frac{2\pi}{T}$$

$$a_k = \frac{1}{T}\int_{-T/2}^{T/2} z(t)e^{-jk\omega t}dt \qquad\qquad b_m = \frac{1}{T}\int_{-T/2}^{T/2} h(t)e^{-jm\omega t}dt$$

  - Linear convolution can produce output = ∞ (as h(t) is unstable & non-causal)
- Periodic convolution: $z(t) \circledast h(t) = \int_{-T/2}^{T/2} z(\tau)h(t-\tau)d\tau = \sum_{k=-\infty}^{\infty} a_k b_k T e^{jwkt}$
  - Fourier series coefficients of periodic convolution output is $a_k b_k T$

Fourier Series – Filter Examples

- FS at the output of a differentiator:
  - $dz(t)/dt \rightarrow a_k\, jkw$
  - High-pass filter

- FS at the output of a integrator:
  - $\int^t z(\tau)d\tau \rightarrow a_k/jkw$
  - Low-pass filter

Fourier Series – Examples

- Find the FS of sinusoids: $\sin(wt)$, $\cos(wt)$, e^{jwt}

Fourier Series – Examples

- Find the FS of the impulse train: $\sum_{n=-\infty}^{\infty} \delta(t - nT)$

Fourier Series – Examples

- Find the frequency response of system with $h(t) = e^{-t}u(t)$ and predict the output of this system for a sinusoid of very low and high frequency
  - Voltage across capacitor in an RC circuit
  - Draw its frequency response as a function of w

- Frequency response of RC circuit $\lambda_{kw} = 1/(1 + RCjkw) \rightarrow$ an LPF

Summary of Fourier Series

Summary

- Phasors are eigenfunctions of LTI systems : orthogonal and bounded.
- Fourier series: Decomposition of a periodic signal in terms of phasors
  - Phasors of a fundamental frequency and its harmonics
  - FS coefficients: Component (projection) of a phasor in the given signal
  - Infinite phasors may be required for representation: Gibbs phenomenon
  - Spectrum and bandwidth depend on the frequency components present
  - Parseval's theorem: Power/energy in any representation are equal

Summary

- Dirichlet conditions ensure existence and convergence of FS
- Output of an LTI system in terms of FS coefficients is the product of FS coefficients of the input and the frequency response of the LTI system
  - Frequency response is the eigen value of the LTI system corresponding to the phasor of the given frequency
- Multiplying two signals results in the (discrete) convolution of their FS coefficients
- For even signals: $z(t) = b_0 + \sum_{k=1}^{\infty} b_k \cos k\omega t$, for odd signals: $z(t) = \sum_{k=1}^{\infty} b_k \sin k\omega t$

Summary

$z(t)$ – Time domain	a_k – Fourier coefficients
Even	Even
Odd	Odd
Conj. Sym.	Real
Conj. Asym.	Imaginary
Real	Conj. Sym.
Imaginary	Conj. Asym.

$z(t)$ – Time domain	a_k – Fourier coefficients	Bandwidth
$z^*(t)$	a_{-k}^*	No change
$z(-t)$	a_{-k}	No change
$z(t-s)$	$a_k e^{-jkws}$ (Only phase changes)	No change
$z(ct)$	$a_{k/c}$ (Coefficients are same but fundamental frequency changes)	c>1: expands c<1: shrinks